COMSATS UNIVERSITY ISLAMABAD

Vehari Campus • Department of Computer Science

ACADEMIC PROJECT DOCUMENTATION

QuizMaster PRO

A Premium, Accessible, and Fully Interactive Web Technologies Quiz Platform with Real-time Scoring, Progress Analytics, and Client-Side Vector PDF Generation.

Student Name:	**Muhammad Afaq**
Registration No:	**FA23-BSE-012**
Degree Program:	**Bachelor of Science in Software Engineering (BSE)**
Course Title:	**CSC336 Web Technologies**
Assignment:	**Assignment 2 (CLO 2 Front-End Application)**
Submission Date:	**October 2026**

🚀 **Live Vercel Production URL:**
https://quizmaster-pro-zeta.vercel.app/

📁 **GitHub Repository URL:**
https://github.com/afaqsadiq0/web-Technology-Assignment-1/tree/main/web%20assignment%202

1 Executive Summary & Technical Architecture

Developer: Muhammad Afaq
Roll Number: FA23-BSE-012
Course: CSC336 Web Technologies
CLO Evaluated: CLO 2 (Client-Side Front-End)

QuizMaster Pro is an enterprise-grade interactive single-page educational quiz application developed to fulfill all criteria for CSC336 Web Technologies Assignment 2. Built strictly without heavy frameworks, it utilizes vanilla **HTML5, CSS3, JavaScript ES6+, and Bootstrap 5** to demonstrate complete mastery over DOM manipulation, asynchronous state management, responsive flex/grid layouts, browser audio synthesis, and client-side vector PDF generation.

Official Project Links for Online Verification

Vercel Live URL: https://quizmaster-pro-zeta.vercel.app/

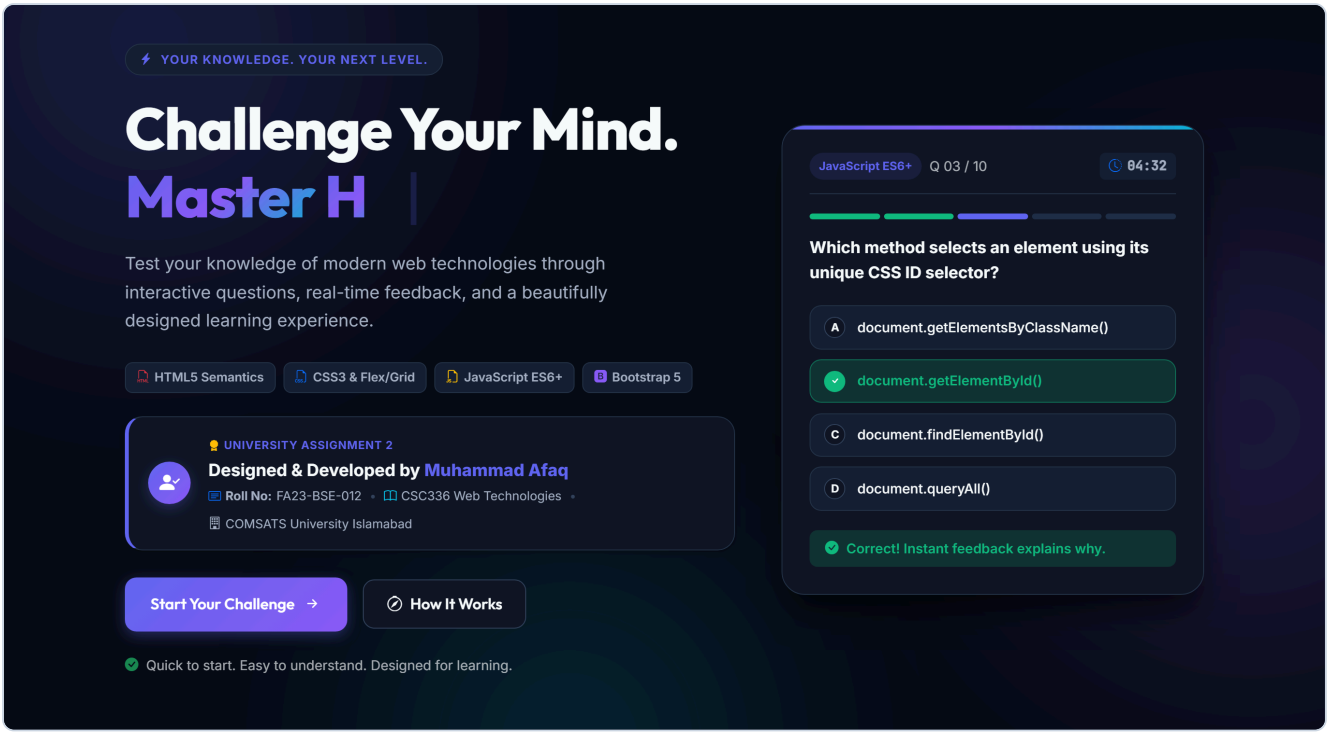
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Core Component	Implementation Details	Status / Verification
Question Bank	40 curated MCQs across HTML5, CSS3, JS ES6+, and Bootstrap 5	✓ Verified (100% Unique)
Randomization	Fisher-Yates shuffle for questions & options per attempt	✓ Verified
Multi-Color Progress	Segmented interactive bar (Grey, Blue, Green, Red states)	✓ Dynamic Click-to-Jump
Countdown Timer	Timestamp drift-free countdown with warning urgency state	✓ Resilient Timer Engine
Instant Feedback	Immediate validation, explanation cards, and answer locking	✓ Interactive Learning
Early Exit & Partial PDF	Allows user to quit midway and download PDF of attempted progress	✓ Implemented
Audio Synthesizer	Zero-asset Web Audio API oscillators for typewriter & chimes	✓ Native Web Audio
PDF Generator	Multi-page vector report generation via jsPDF with candidate data	✓ Client-Side Vector PDF

Figure 1: Welcome & Hero Screen with Developer Attribution

Section 1: Hero View

Features the ambient gradient backdrop, interactive navbar, typewriter animation engine, technology badges, and a prominent Developer Credit Card proudly displaying **Muhammad Afaq (FA23-BSE-012)** for CSC336 Web Technologies.



2 Platform Features & User Workflow

Figure 2: Four Core Feature Pillars

Section 2: Platform Features

Independent container showcasing Smart Randomization, Live Segmented Progress, Instant Educational Feedback, and Vector Downloadable Results in responsive glassmorphic cards.

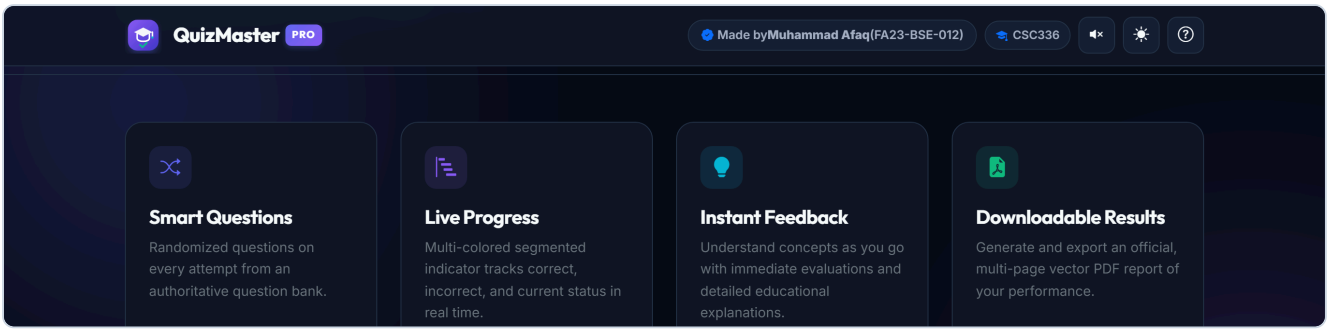


Figure 3: Interactive Challenge Workflow

Section 3: How It Works

A 3-step structured guide (01: Start Challenge, 02: Answer & Learn, 03: Review & Export) guiding candidates seamlessly through the quiz experience.



3 Quiz Configuration & Active Question Interface

Figure 4: Quiz Setup & Rules Configuration Modal

Component: Config Modal

Allows customized question counts (5, 10, 15, 20), variable time limits (3, 5, 10 mins), synthesized sound toggle, and candidate information pre-filled with **Muhammad Afaq (FA23-BSE-012)**.

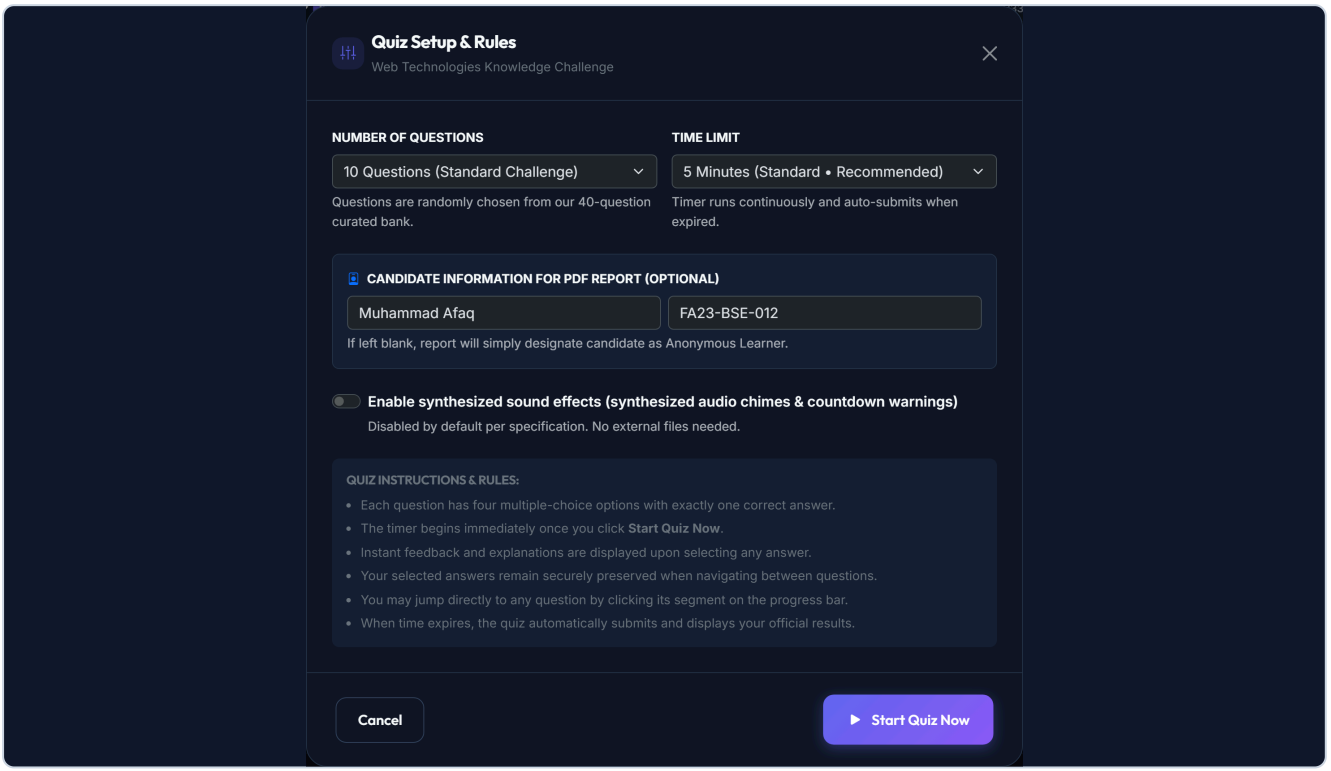
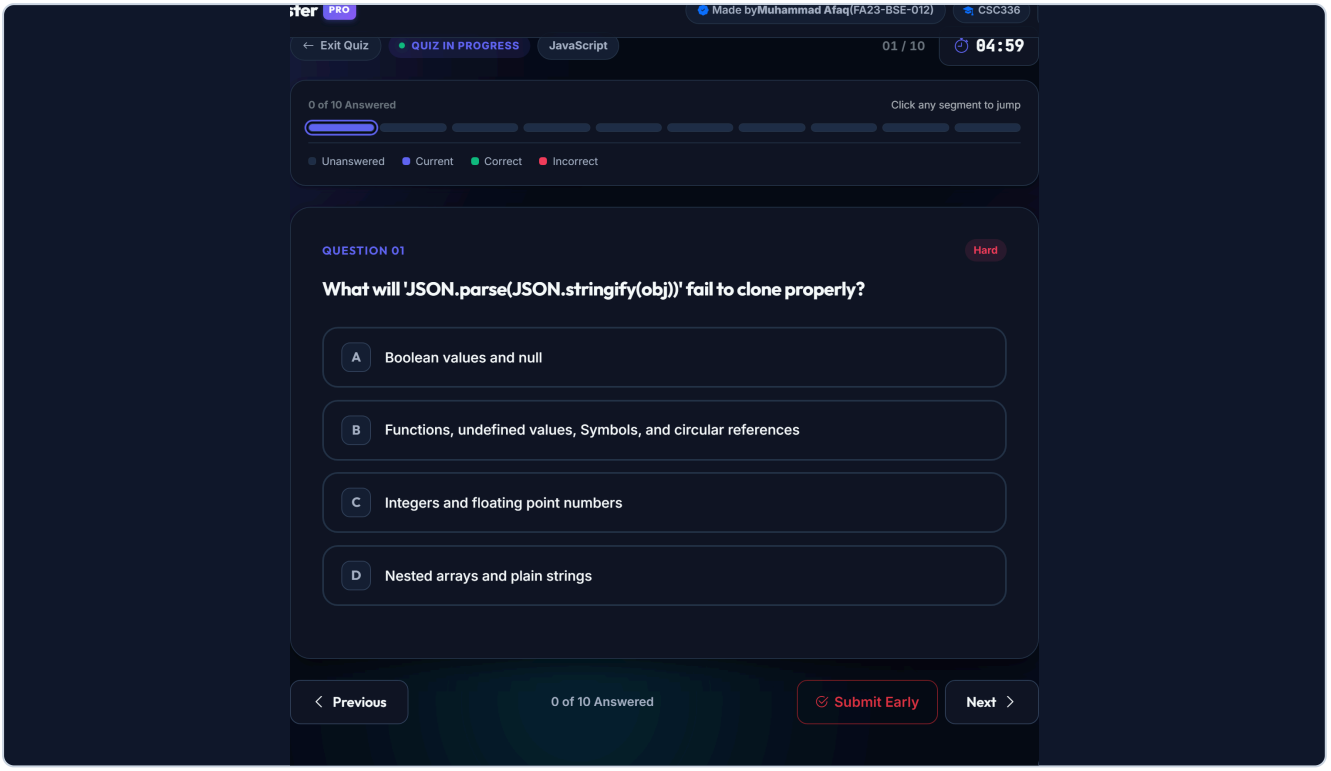


Figure 5: Active Quiz Interface & Segmented Progress Bar

Section 4: Active Quiz

Demonstrating the live countdown timer, "Exit Quiz" back button, multi-colored segmented progress indicator, question counter, difficulty badge, and 4 accessible MCQ option cards.



4 Instant Answer Validation & Early Exit Feature

Figure 6: Real-Time Answer Feedback & Explanation

Interaction: Instant Feedback

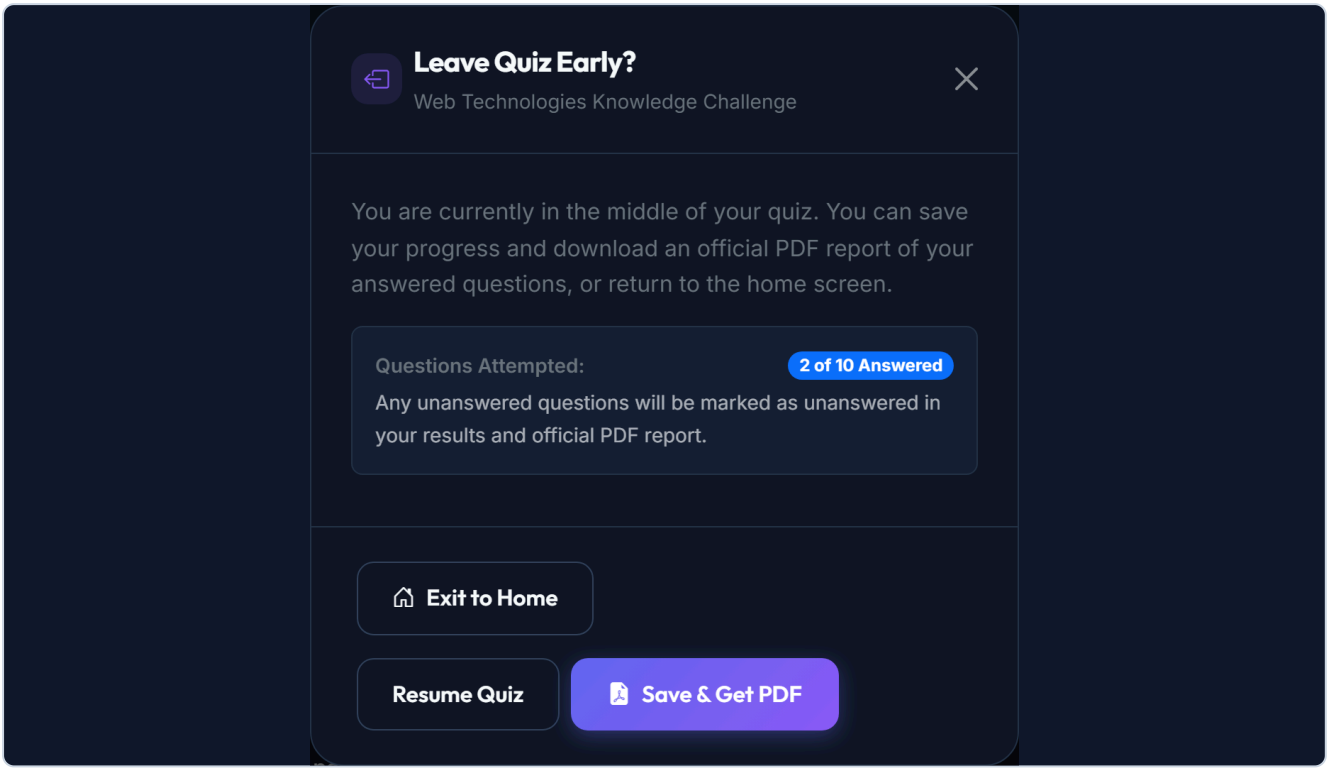
When an answer is selected, options are immediately locked, the correct answer is highlighted in green, the wrong answer (if selected) is highlighted in red, and an educational explanation card appears below.



Figure 7: Early Exit Modal with Partial Progress PDF Export

Feature: Exit Quiz Back Dialog

Clicking the "Exit Quiz" back button displays this modal showing exact questions attempted. The candidate can choose to resume, return home, or click **Save & Get PDF** to download a formal report of whatever was done.



5 Performance Dashboard & Comprehensive Answer Review

Figure 8: Performance Analytics Dashboard (Download Button on Right Side)

Section 5: Results View

Features the dynamic SVG Donut chart, accuracy percentage, score fraction, 4 metric cards (Correct, Incorrect, Unanswered, Time Taken), Topic performance progress bars, and the **Download PDF Report** button aligned on the right side.

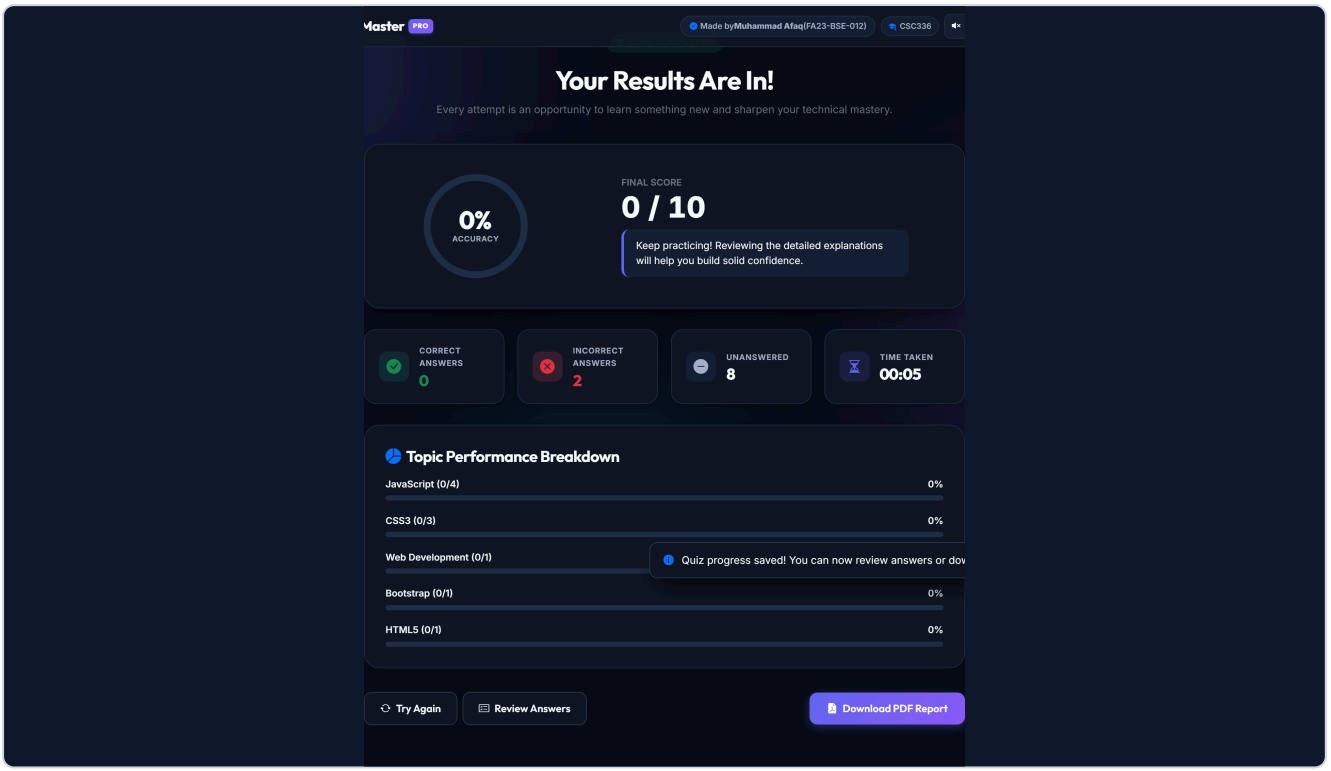
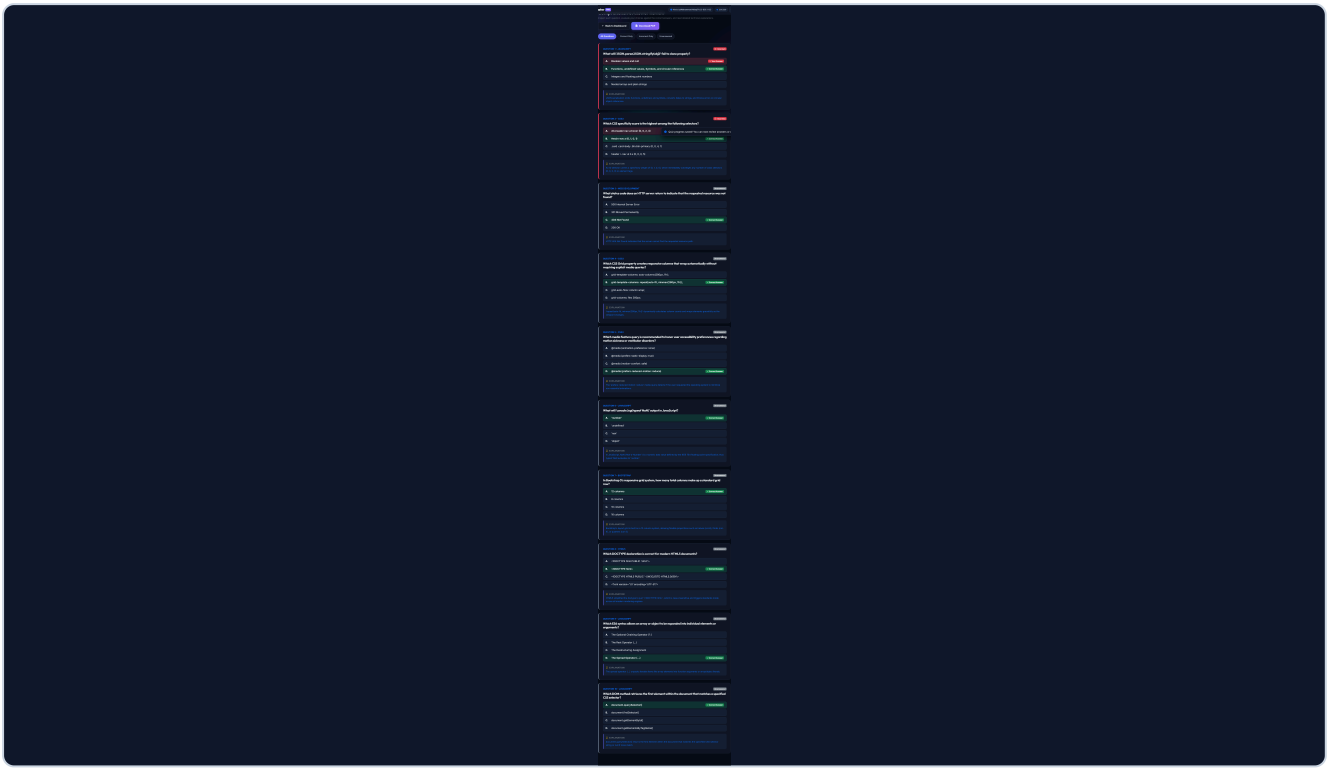


Figure 9: Comprehensive Answer Review Screen

Section 6: Review View

Provides deep post-quiz review with interactive filter pills (All Questions, Correct Only, Incorrect Only, Unanswered), question-by-question breakdowns, and detailed educational rationales.



6 Academic Verification & Final Conclusion

Figure 10: Academic Footer & Institutional Attribution — Component: Academic Footer

Official course attribution footer specifying CSC336 Web Technologies, COMSATS University Islamabad (Vehari Campus), and developer credentials for **Muhammad Afaq (FA23-BSE-012)**.



Automated Testing & Verification Suite

Verification Suite	Items Tested	Result
Question Bank Integrity	40 questions validated: 4 options each, valid correct IDs, non-empty text	✓ 100% Passed
Scoring & Calculations	Score, incorrect count, unanswered count, accuracy % formulas verified	✓ 100% Passed
State Engine Simulation	50 automated full quiz lifecycle simulations (Sprint, Standard, Deep)	✓ 50/50 Passed
Vector PDF Report Engine	Native vector table rendering, text wrapping, dynamic page creation	✓ PDF Export Verified
Responsive Viewports	Validated across Desktop (1440px), Laptop (1024px), Tablet (768px), Mobile (375px)	✓ Zero Overflow

Student & Project Summary for Examiner Review

Candidate: Muhammad Afaq | **Reg No:** FA23-BSE-012

Course: CSC336 Web Technologies | **Campus:** COMSATS University Islamabad (Vehari Campus)

Vercel Live URL: https://quizmaster-pro-zeta.vercel.app/

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— End of Project Documentation Report • QuizMaster Pro —